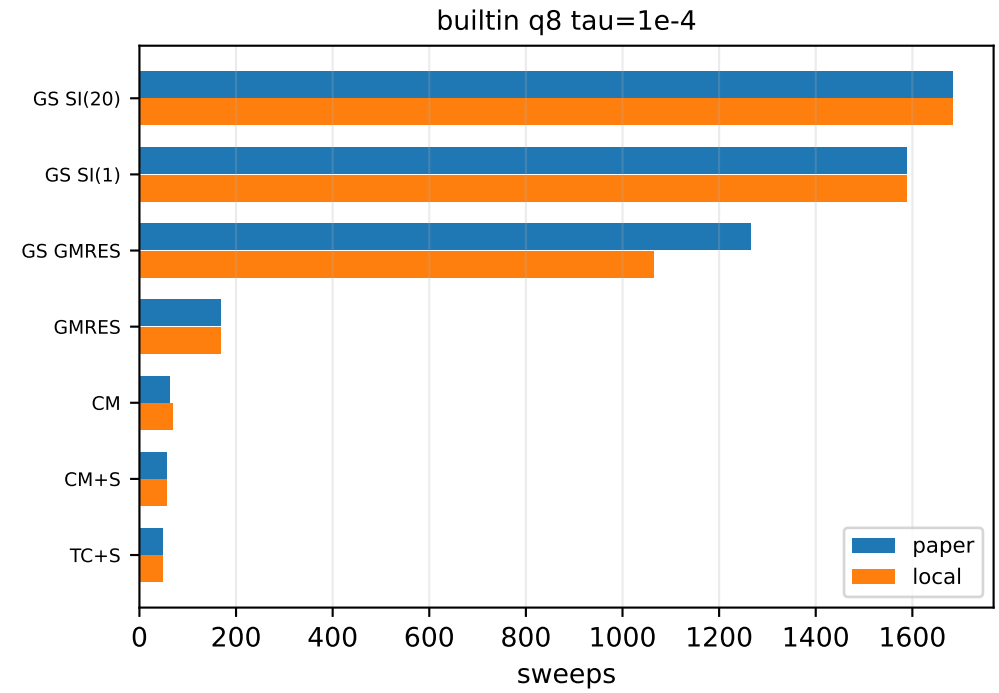
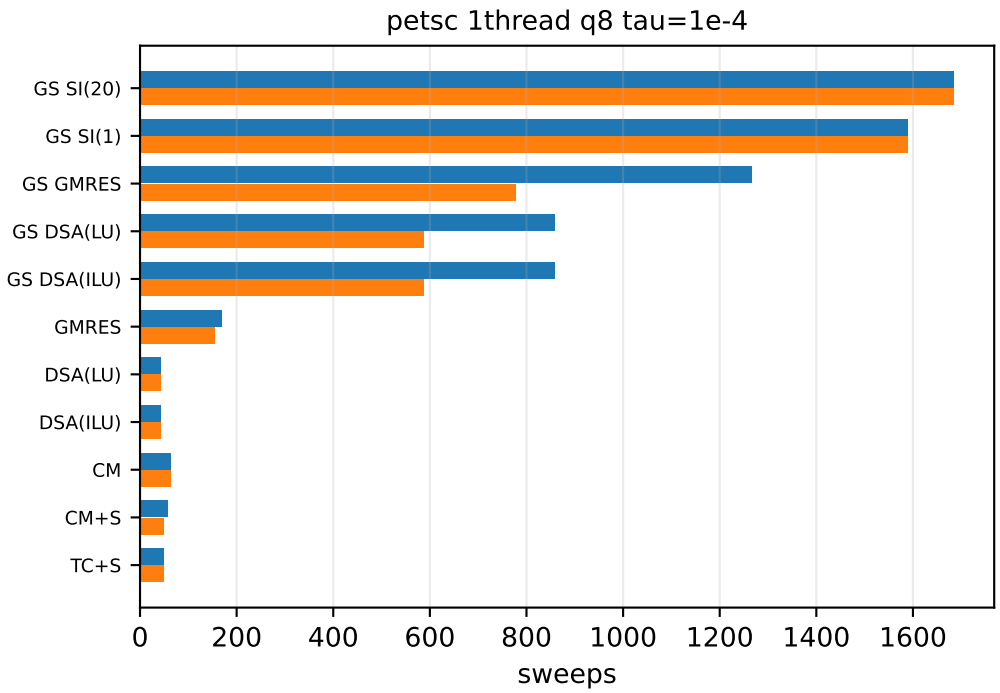
CMDSA Python Example: Updated Local Results

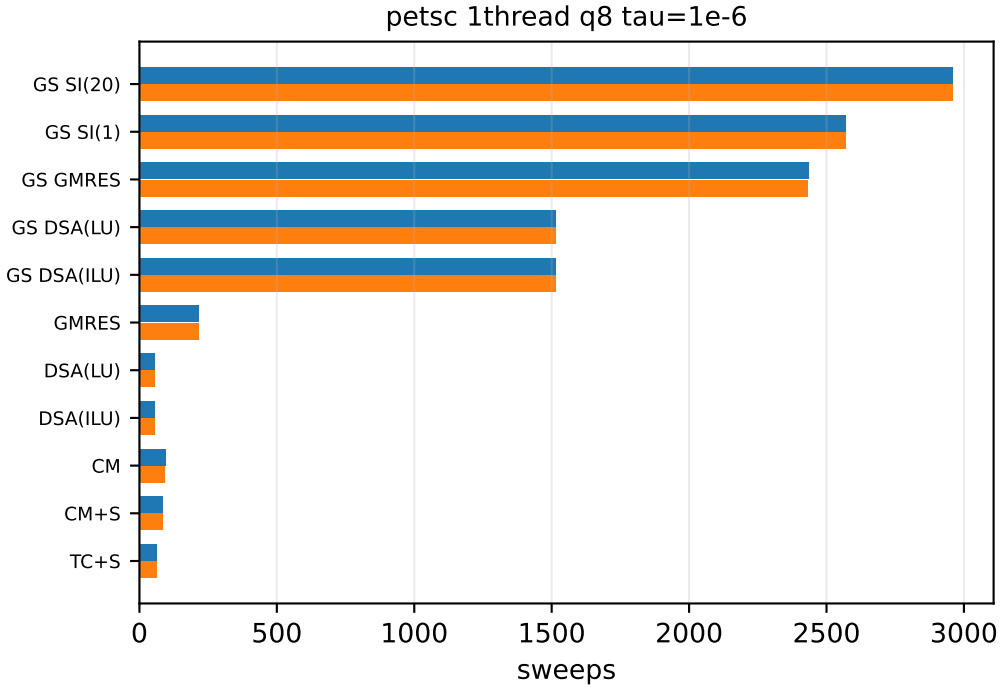
Multigroup Diffusion Preconditioners for Multiplying Fixed-Source Transport Problems
Jeremy A. Roberts and Benoit Forget, JCP 274 (2014), pp. 455-472
https://doi.org/10.1016/j.jcp.2014.06.034

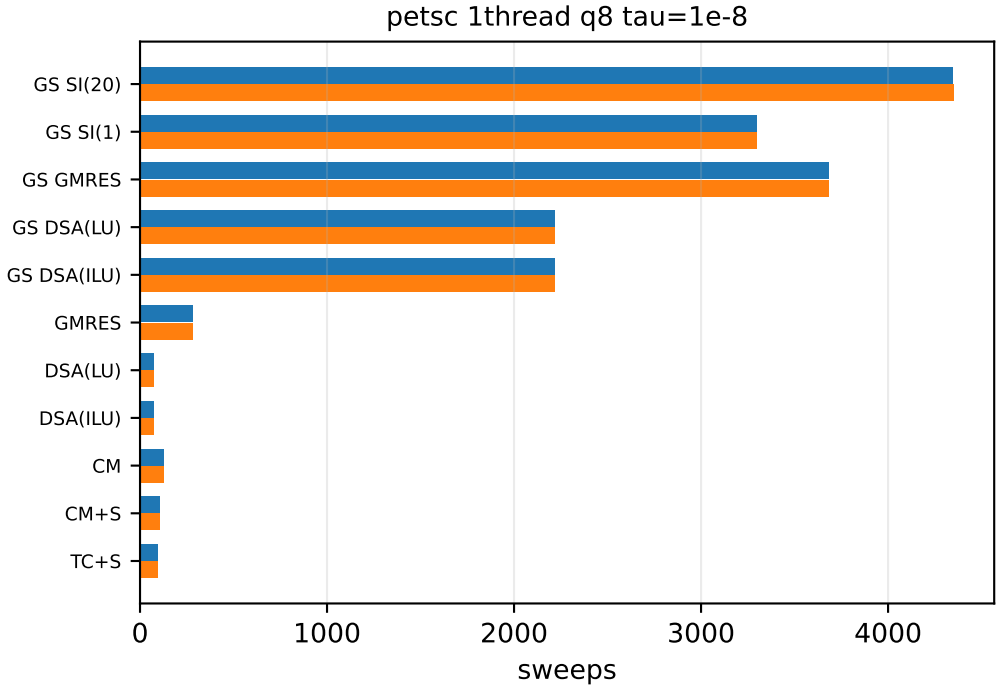
- Archive source: /home/robertsj/Research/detran/detran-examples/mgcmdsa/updated_results
- C5G7 rows are local q8 timing/sweep runs compared with the recovered paper tables.
- BWR spectral rows use corrected fission material data for A/DSA/CMDSA/CMDSA-S.
- Corrected-fission TC-CMDSA was not regenerated in this pass.
- Transport nnz uses the archive drop tolerance; tiny entries near 1e-9 explain most nnz drift.

C5G7 Sweep Counts

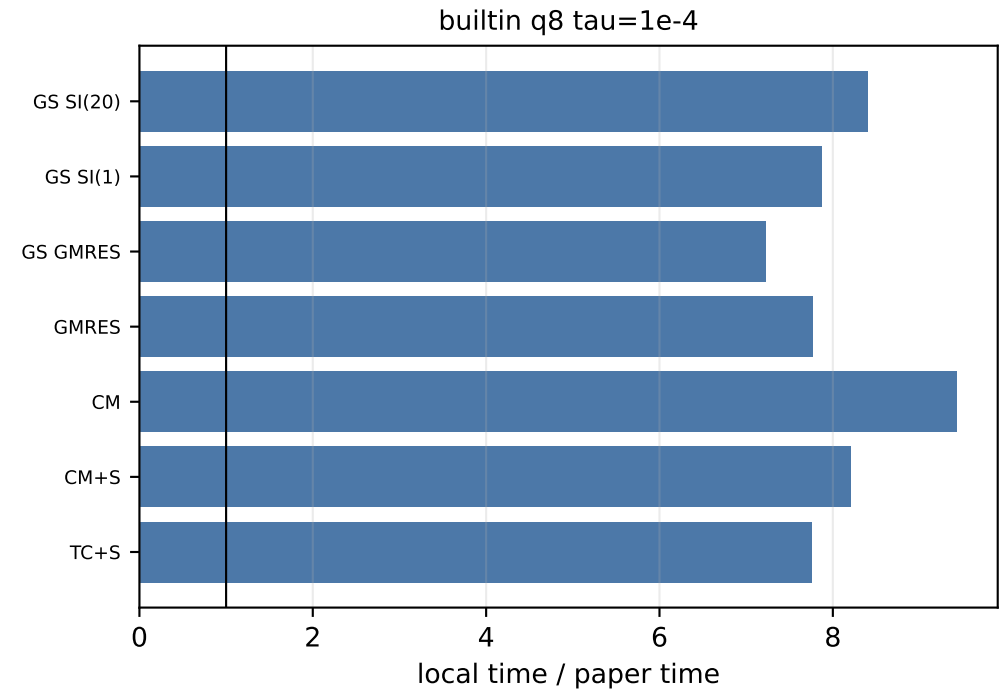


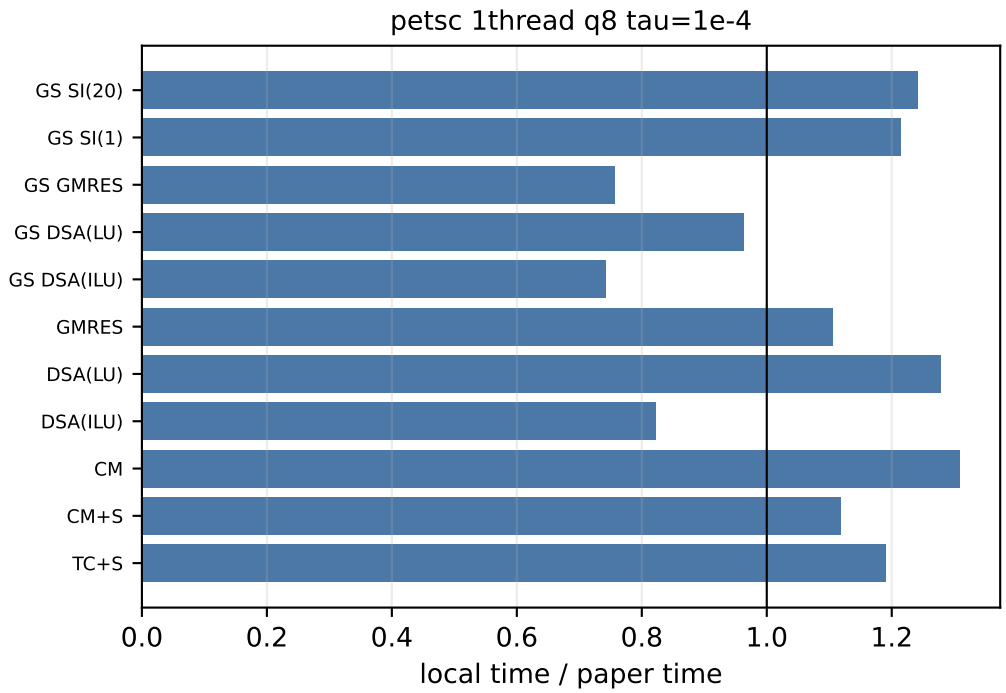


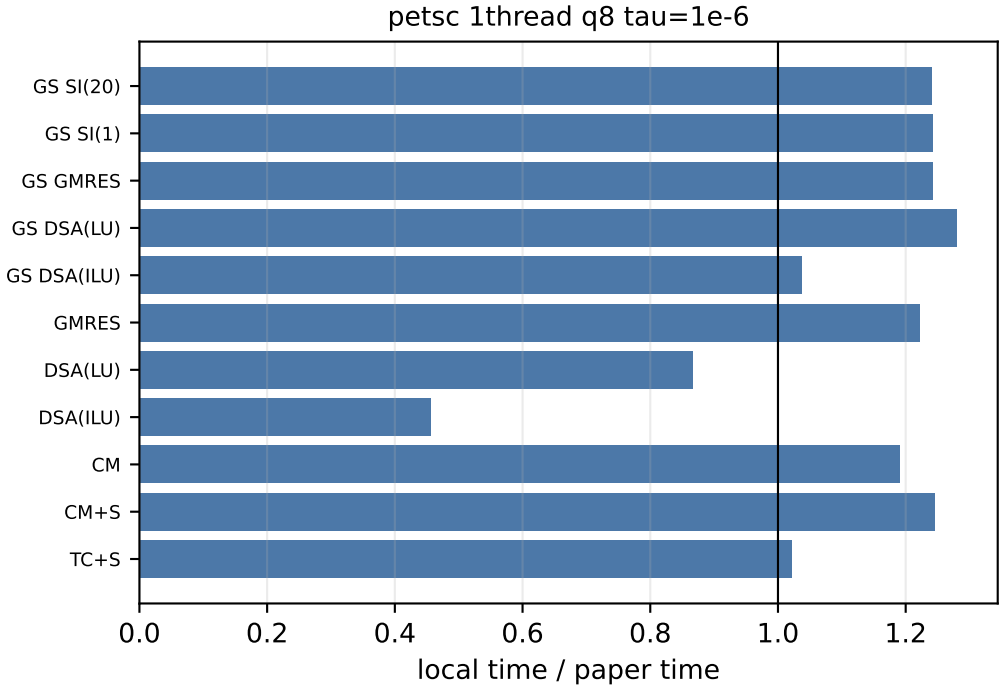


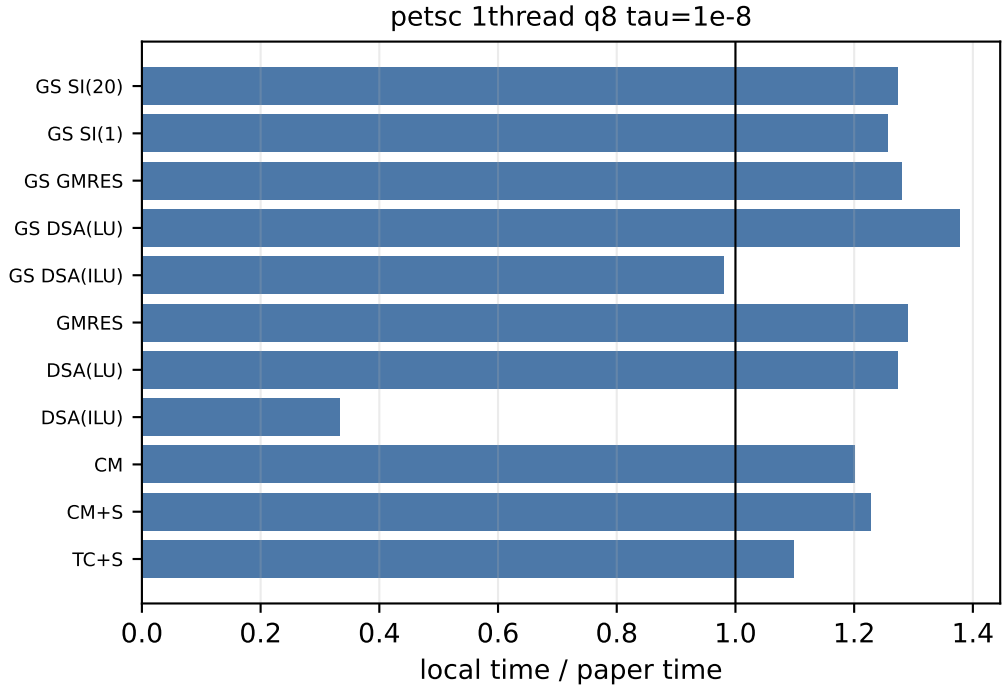


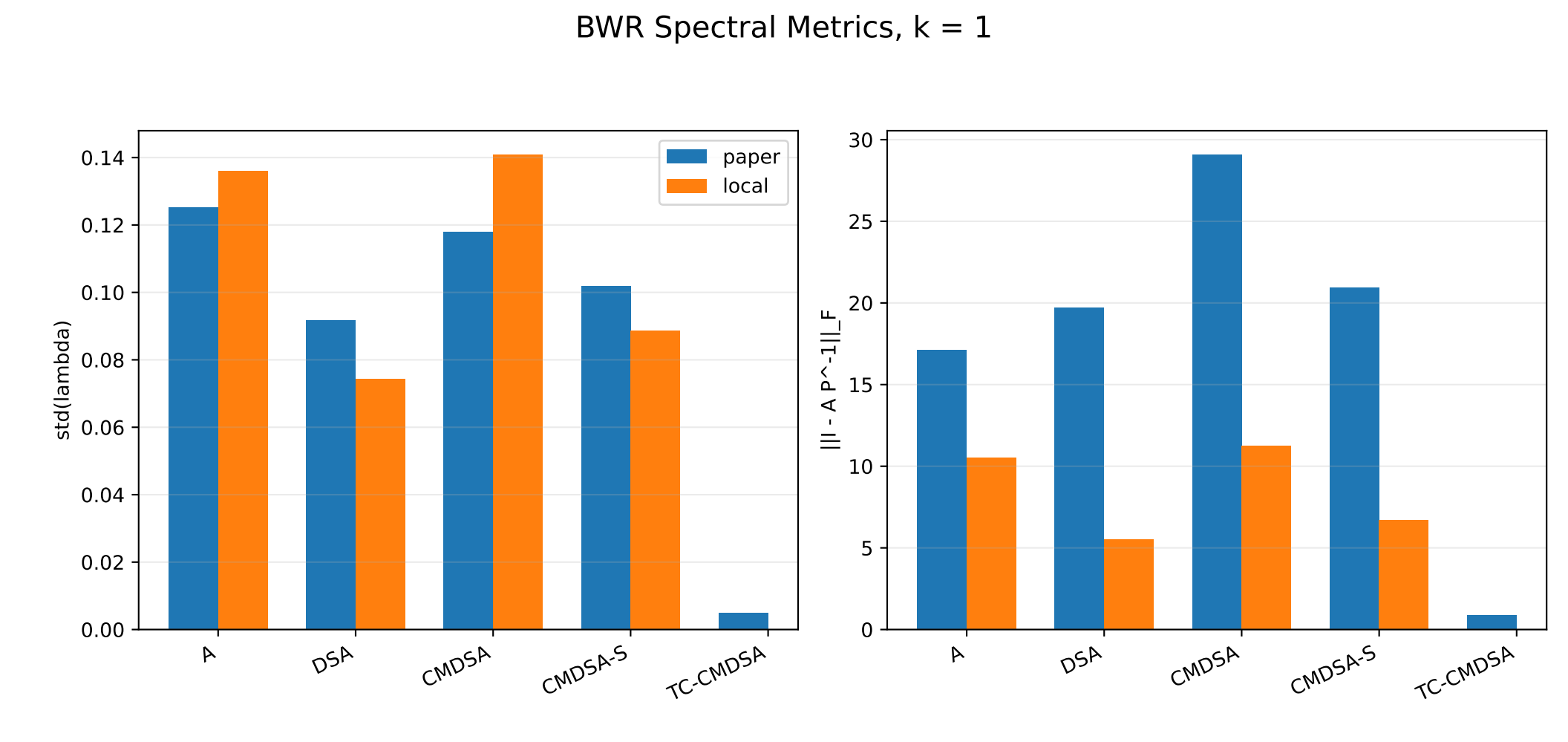
C5G7 Timing Ratios










BWR Spectral Metrics, k = 1
std(lambda)
0.00
0.02
0.04
0.06
0.08
0.10
0.12
0.14
paper
local
A
DSA
CMDSA
CMDSA-S
TC-CMDSA
||I - A P^-1||_F
0
5
10
15
20
25
30
A
DSA
CMDSA
CMDSA-S
TC-CMDSA

BWR Eigenvalue Spectra From Archived Local Run

